What is it?

As time goes by the amount of data that flies invisibly between our phones is increasing. In 1996 we saw the introduction of SMS and voice in India. Soon we had far richer data capabilities — multimedia and comfortable text surfing on the net. In the future, we're going to be seeing high-definition video fly in and out of devices the size of a handset. We're talking dozens of megabytes per second here. Theoretically UMB (Ultra Mobile Broadband) would allow you to handle 275 megabytes per second download and 75 MBps upload, even in a car travelling at 60 km per hour. When the Indian telecom industry started out about 14 years ago, we had to struggle with call-drops while standing still. There's still enormous scope for improvement. It's not just because of improving handsets. It's also because of what's happening at the network end.

However, Telecom Engineering is a lot larger than just mobile phones. This is just the most visible part as a consumer. This field is larger than telephones. It's about communications. The entire industry is dedicated to ensure that individuals, organisations and even continents can communicate more efficiently.

Broadly put, telecommunications is about the transfer of data. We however, are most familiar with that chunk of plastic and metal in our pockets. Beyond that we have networks, satellites and undersea cables.

Why is it needed?

With the explosion of electronic media, telecommunication engineering/ infrastructure is one field that is gaining immense popularity as a lucrative career option for the new generation.

As you drive through the city you see huge towers everywhere. Each of these is busy beaming signals across its circle to ensure you can stay connected. There is a huge infrastructure that goes behind a telecom company. This includes the setting up of towers, laying of cables where necessary and then bringing the whole customer end together. Basically, a cell site consists of electronic (active) and non-electronic infrastructure (further defined below):

- Electronic infrastructure includes base tower station, microwave radio equipment, switches, antennae, transceivers for signal processing and transmission.
- Non-electronic infrastructure includes tower, shelter, air-conditioning equipment, diesel electric generator, battery, electrical supply, technical premises and easements and pylons that account for nearly 60 per cent of network roll-out costs.

Telecom is a mammoth industry and opens a wide range of career opportunities on both the hardware and software fronts. The list is long: mobile telephony, internet protocol media system, wireless communications, GSM (Global System for Mobile Communications), GPRS (General Packet Radio Service), CDMA (Code Division Multiple Access), VoIP (Voice over Internet Protocol), data networks, optical networks, and much more. Internet Service Providers use telecom hardware and therefore employ telecom engineers for their network-related operations.

The growth story

The teledensity was 34.5 per cent as in Jan 2009 and the projected teledensity is 500 million, 40 per cent of population by 2010. One can just imagine the number of people who will be required to service the demands of such a fast